(a) Physical intervention

Load setpoint



$$(P, Q) \mapsto (1 + a)(P, Q)$$

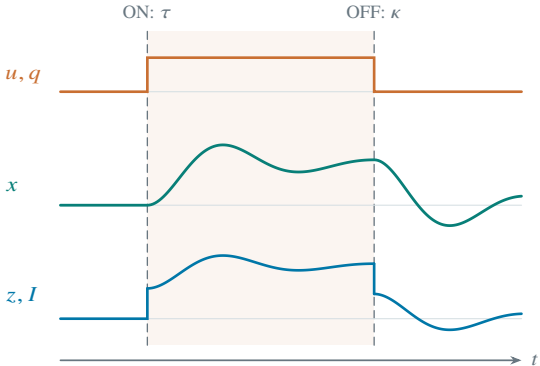
Line opening



$$q_0 \mapsto q_{\text{open}}$$

$$\dot{x} = f_q(x, z, u)$$
$$0 = g_q(x, z, u)$$

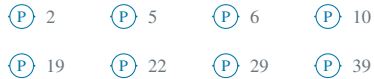
(b) Restoration and recovery



$$x(\kappa^+) = x(\kappa^-) \quad g_{q_0}(x(\kappa), z(\kappa^+), u_0) = 0$$

Illustrative response; amplitudes and time are arbitrary.

(c) Reported measurements



Voltage sites; current terminals mapped separately

$$\text{Phasors } h_k = H_q(x_k, z_k, u_k)$$
$$\text{Phase history } \eta_k = F_M(\eta_{k-1}, h_k)$$



$y_k \in \mathbb{R}^{47}$ at 30 reports/s

BUS39 current angle excluded